The Gas Laws

Boyle, Charles, Gay-Lussac, the combined gas law and PV = nRT

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS

$P_1V_1 = P_2V_2 \cdot \frac{V_1}{T_1} = \frac{V_2}{T_2} \cdot \frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \cdot PV = nRT$

$R = 0.08206$ L·atm/(mol·K) $= 8.314$ L·kPa/(mol·K)

K = °C + 273.15 · Molar masses: C 12.01, O 16.00, Ar 39.95

Give answers to three significant figures.

A. Understanding

1. Explain, in terms of particles, why the pressure of a gas increases when its volume is reduced at constant temperature.

2. Why must temperatures be converted to kelvin in gas law calculations?

B. Simple gas laws

3. A gas occupies 5.0 L at 100 kPa. What is its volume at 250 kPa, at the same temperature?

4. A balloon contains 250 mL of air at 20 °C. It is cooled to −30 °C at constant pressure. What is its new volume?

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5. A car tyre is at 220 kPa at 15 °C. After a long drive, the air inside reaches 45 °C. Assuming the volume stays constant, what is the new pressure?

C. Combined gas law

6. A 4.00 L sample of gas at 1.00 atm and 25 °C is changed to 2.50 atm and 100 °C. What is its new volume?

D. Ideal gas law

7. Calculate the pressure, in atm, of 0.250 mol of gas in a 5.00 L container at 300 K.

8. A 2.00 L flask contains carbon dioxide at 1.50 atm and 27 °C. Calculate (a) the number of moles of CO_2 and (b) its mass.

E. Challenge

9. A 1.00 g sample of a gas occupies 0.560 L at 1.00 atm and 0 °C. Calculate its molar mass and suggest which noble gas it is.

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